

Internship at LSI Industries Summary

Date: Summer 2026

Role: Engineering Intern

Objective: A comprehensive breakdown of technical contributions, engineering disciplines, and operational processes learned during the internship.

1. Mechanical Engineering

Applied core mechanical design principles using Autodesk Inventor to design components, create manufacturing drawings, and refine assemblies.

- **LAW2 End Cap Redesign:** Redesigned die cast aluminum end caps to incorporate a physical stop that prevents gasket over-compression. This involved detailing draft angles, tolerance stacks, and physical clearances.
- **CNC Subplate Design:** Redesigned a heavy-duty subplate for CNC machining fixtures. This work focused on developing precise drawings, identifying datums, and ensuring accurate positioning.
- **Sheet Metal Design:** Modeled a 4-sided light shield in Inventor. Used sheet metal manufacturing standards to calculate bend allowances and generate flat-pattern drawings with fold-lines.

2. Electrical Engineering

Gained practical, hands-on exposure to electrical sub-assemblies and wiring schematics.

- **Velocity (LSI product) Family Modules:** Hands-on assembly of several power and control modules from the Velocity product family.
- **Systems Integration:** Studied electrical schematics to understand how individual electronic components interface and function together to achieve optimal performance.
- **Skills Developed:** Overcame a previous hesitation toward electrical engineering by building confidence in hardware assembly, wiring, and circuit pathing.

3. Research & Development (R&D) & Product Development

Supported the product development lifecycle through iterative design, physical testing, and prototype validation.

- **Iterative Optimization:** Managed **11 design iterations** of the LAW2 end cap, systematically adjusting CAD models based on feedback regarding manufacturing constraints, casting parameters, and part costs.
- **Rapid Prototyping:** Utilized **3D-printing** to prototype and test iterations of the end cap, validating physical tolerances and fit-up before final production tooling was cut.

- **Physical Verification (Mount Test):** Assisted with physical mount testing. This highlighted the real-world discrepancies between perfect CAD models and actual physical assemblies, emphasizing the importance of mock-up testing.

4. Manufacturing Engineering

Worked directly with manufacturing processes and constraints to ensure designs are optimized for production (Design for Manufacturability).

- **Shop Floor Operations:** Spent time in the pole plant and packaging plant picking parts and painting fixtures. This hands-on experience provided direct visibility into the downstream manufacturing flow.
- **Casting Constraints:** Learned draft angle standards, casting tolerances, thermal shrinkage allowances during the design of cast components, and more.
- **Drawing Standards:** Created production drawings using industry-standard symbols, clear views, and standard sheet metal notes.

5. Design Automation & Process Engineering

Designed automation workflows and created systems to standardize repetitive engineering tasks.

- **iLogic/Code Development (The Pole Generator):** Developed a parameter-driven pole generation tool in Autodesk Inventor using **8,000+ lines of custom iLogic code**, created in collaboration with AI tools.
- **Data Integrity (PDM/ERP Integration):** Connected the automation system with **Autodesk Vault** for document control and **JD Edwards** for ERP/BOM synchronization.
- **Process Standardization (SOP):** Authored a Standard Operating Procedure explaining how the generator works, its backend architecture, and maintenance guidelines. Presented this documentation to the team to ensure a clean transition of ownership.

6. Operations & Administration

Learned how engineering departments coordinate projects, manage changes, and interact with the broader business.

- **Engineering Change Notices (ECN):** Performed ECN reviews, learning how to document revisions properly and query change logs to trace why historic designs were altered.
- **Project Management Shadowing:** Shadowed a project manager to learn about project scoping, team scheduling, and how PMs align production and engineering resources.
- **Agile Standups:** Participated in morning team meetings to track project paths, discuss obstacles, and practice speaking and presenting to a team of senior engineers.
- **Vendor Relations:** Attended technical "Lunch-and-Learn" seminars with external suppliers, gaining insight into the procurement and business dynamics of engineering.